

NetEngine 8000 F8 Service Router

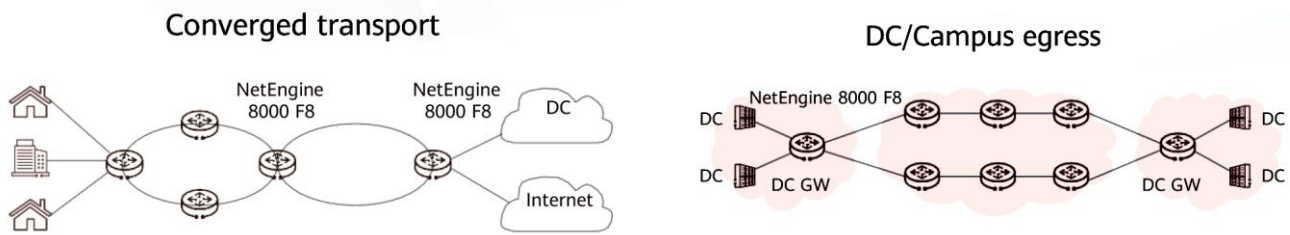
Overview

The NetEngine 8000 F8 is the ultra-high-density multi-service aggregation Router. It only 13U high and 482.8 mm deep, it provides a port capacity of up to 2 Tbps. Moreover, it provides eight service slots, supporting combined line processing units and physical interface cards designed by parent-child cards to flexibly adapt to various services. It supports 100GE, 50GE, 10GE, GE, FE, POS, CPOS, and E1 interfaces. Its key hardware components support redundancy backup, achieving high reliability. And with features such as SRv6/EVPN, FlexE, and 1588v2, the NetEngine 8000 F8 is an ideal choice for converged multi-service transport in the cloud era.



NetEngine 8000 F8 service router

Usage Scenarios



Product Specifications

Specification	Description
Dimensions (H x W x D)	578.5 mm (13U) x 442 mm x 482.8 mm
Forwarding capacity	2 Tbps
Voltage range	DC: -40 V to -72 V AC: 90 V to 290 V
Typical power consumption	1680 W
Maximum number of slots	8, supporting parent-child cards and combined line processing units, up to 32 PICs, 8 LPUCs
Maximum capacity per slot	400G
Maximum interface capacity per device	24 x 100GE; 240 x 10GE; 320 x GE; 512 x E1; 256 x STM; 512 x E1
Operating temperature	Long term: 0°C to +45°C Short term: -5°C to +55°C
Operating humidity	Short term: 5% RH to 95% RH, non-condensing Long term: 5% RH to 85% RH, non-condensing

Product Features

Feature	Description
L2 function	IEEE 802.1q, IEEE 802.1p, IEEE 802.3ad, IEEE 802.1ab, STP/RSTP/MSTP
L3 function	OSPF, RIP, IS-IS, BGP, ACL, 6VPE, ARP, VLANIF, VXLAN, EVPN
Routing	Routing protocols such as RIP, RIPng, OSPF, OSPFv3, IS-IS, IS-ISv6, BGP, BGPv6, static routing, IPv4 multicast, IPv6 multicast, and static multicast TCP/IP protocol family, such as ICMP, IP, TCP, UDP, socket (TCP/UDP/RAW IP), and ARP Static DNS, DNS Client, FTP Server, FTP Client, TFTP Client DHCP relay and DHCP server
Network slicing	FlexE slicing, minimum 1G FlexE granularity, channelized sub-interface, and network slicing based on slice IDs

Feature	Description
Service visibility	Flow-based detection IFIT
MPLS	LDP, RSVP-TE, L2VPN, L3VPN, Seamless MPLS
Segment Routing	SR BE, SR Policy, SRv6 BE, SRv6 Policy, EVPN L3VPN, EVPN VPWS, EVPN VPLS, SRv6 FlexAlgo, G-SRv6
Multicast	IGMP, Static Multicast Routing, PIM-SM/SSM, MBGP, BIERv6, BIER MPLS, MVPN over BIERv6, NG MVPN over BIER
Clock	1588v2, 1588 ATR, 1588 ACR, CES ACR, synchronous Ethernet, extended SSM
NAT	NAT44, NAT ALG, NAT internal server, NATPT, No-PAT
Security	IPsec, ARP, BGP FlowSpec, DHCP, URPF, 802.1x, loop detection, MACsec, VLAN-IN-Clear, CMPv2, IKEv2, GRE over IPsec, NAT traversal, IPsec VPN, Keepalive and DPD for Peer Detection and Dynamic Remote IPsec Access
IPv4	Tracert, NQA, IP policy-based routing, specified next hop for forwarding based on flows, IP policy-based routing load balancing, QinQ interfaces (QinQ and dot1q termination), IPv4 load balancing, and enabling/disabling the fast reply function based on the interface board, egress of the MPLS in UDP tunnel
IPv6	IPv6 (ND), Path MTU (PMTU), TCP6, Ping IPv6, Tracert IPv6, Socket IPv6, DHCPv6 Relay, Static IPv6 DNS, TFTP IPv6 Client, IPv6 Policy Routing, IPv6 ND fast reply
OAM	IFIT, IP FPM, NQA, TWAMP, BFD, MPLS OAM, MPLS-TP OAM, VRRP, Ethernet OAM, 802.3ah, Y.1731, Ethernet LPT, bit-error-triggered switching
QoS	QPPB, DiffServ, HQoS, redirection

Basic Hardware Configurations

NetEngine 8000 F8 Basic Configurations

Order Name	Description	Remarks
CR8PF8BASDC0	NetEngine 8000 F8 Basic Configuration (Includes NetEngine 8000 F8 Chassis, 2*MPUB1, 2*SRU-2T-A, 3 DC Power, 2 Fan Tray, without Software Charge and Document)	
CR8PF8BASAF2	NetEngine 8000 F8 Basic Configuration (Includes NetEngine 8000 F8 Chassis, 2*MPUB1, 2*SRU-2T-A, 2 AC Power, 2 Fan Tray, without Software Charge and Document)	

Interface Cards

Order Name	Description	Remarks
CR8DLPUTK0C0	Trunk Line Processing Unit(LPUT-800-CM, 400G Enabled by Default)	Each slot supports one LPUT and four PICs

Order Name	Description	Remarks
CR8D00LAXFD0	10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC)	
CR8DE2V1NBD0	2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC)	
CR8D00EAGFD5	10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC)	
CR8D00CDE1D0	16-Port E1 Physical Interface Card(PIC,75ohm/120ohm)	
CR8D00C4CFD0	4-Port Channelized STM-1c POS-SFP Flexible Interface Card(PIC)	
CR8D00E8GED0	8-Port 100/1000Base-RJ45 Physical Interface Card(PIC)	
CR8D00P8CFD0	8-Port STM-1c/8-Port STM-4c POS Card (PIC)(Default Enable 4*STM-1c)	
CR8D00E4XFD0	4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC)	

Combined Line Processing Unit

Order Name	Description	Remarks
CR8DERGF00F1	72-port GE(100/1000Base-X-SFP) MACsec Combined Line Processing Unit(LPUC-100)	Each slot supports one LPUC

License

Order Name	Description	Remarks
N1-N8KF8F-LIC	N1-NetEngine 8000 F8 Foundation SW License	
N1-N8KF8A-LIC	N1-NetEngine 8000 F8 Advanced SW License	
N1-N8KF8F2A-LIC	N1-NetEngine 8000 F8 Foundation to Advanced SW License	
N1-N8KF8IFTA-LIC	N1-NetEngine 8000 F8 iFIT Enhanced Function License	
N1-N8KF8IFIT1K-LIC	N1-NetEngine 8000 F8 Series iFIT Flow Scales License(1K Flows)	
N1-N8KF8OAM-LIC	N1-NetEngine 8000 F8 Smart OAM Function License	

Order Name	Description	Remarks
N1-N8KF8TMSY-LIC	N1-NetEngine 8000 F8 Time Synchronization Function License	
N1-N8KF8SRV6-LIC	N1-NetEngine 8000 F8 SRv6 Function License	
N1-N8KF8SLI-LIC	N1-NetEngine 8000 F8 Port Slicing Function License	
N1-N8KF8MACS-LIC	N1-NetEngine 8000 F8 MACSec Function License	
N1-N8KF8IPS-LIC	N1-NetEngine 8000 F8 IPSec Function License	
N1-N8KF8BNG-LIC	N1-NetEngine 8000 F8 BNG Function License	
N1-N8KF8S1K-LIC	N1-NetEngine 8000 F8 Subscribers Quantity(1k Subscribers)	
N1-N8KF8MED-LIC	N1-NetEngine 8000 F8 Media Fabric Function License	
N1-N8KF8SRV6D1-LIC	N1-NetEngine 8000 F8 G-SRV6 32bit SRH Compression Function License	
N1-N8KF8SRV6D2-LIC	N1-NetEngine 8000 F8 G-SRV6 16bit SRH Compression Function License	
N1-N8KF8ESF1D1-LIC	N1-NetEngine 8000 F8 Energy-Saving Enhanced Function Package 1.0	
N1-N8KF8ZTA-LIC	N1-NetEngine 8000 F8 Zero Trust Access Function License	
N1-N8KF8HIPS-LIC	N1-NetEngine 8000 F8 Host-based Intrusion Prevention System Function License	
N1-N8KF8CC-LIC	N1-NetEngine 8000 F8 Configuration Check Function License	
N1-N8KF8TEE-LIC	N1-NetEngine 8000 F8 Security Enhanced Trusted Execution Environment Function License	
N1-N8KF8F-SNS1Y	N1-NetEngine 8000 F8 Foundation SW License SnS	
N1-N8KF8A-SNS1Y	N1-NetEngine 8000 F8 Advanced SW License SnS	
N1-N8KF8F2A-SNS1Y	N1-NetEngine 8000 F8 Foundation to Advanced SW License SnS	
N1-N8KF8IFTA-	N1-NetEngine 8000 F8 IFIT Enhanced Function	

Order Name	Description	Remarks
SNS1Y	License SnS	
N1-N8KF8IFIT1K-SNS1Y	N1-NetEngine 8000 F8 IFIT Flow Scales License(1K Flows) SnS	
N1-N8KF8OAM-SNS1Y	N1-NetEngine 8000 F8 Smart OAM Function License SnS	
N1-N8KF8TMSY-SNS1Y	N1-NetEngine 8000 F8 Time Synchronization Function License SnS	
N1-N8KF8SRV6-SNS1Y	N1-NetEngine 8000 F8 SRv6 Function License SnS	
N1-N8KF8SLI-SNS1Y	N1-NetEngine 8000 F8 Port Slicing Function License SnS	
N1-N8KF8MACS-SNS1Y	N1-NetEngine 8000 F8 MACSec Function License SnS	
N1-N8KF8IPS-SNS1Y	N1-NetEngine 8000 F8 IPSec Function License SnS	
N1-N8KF8BNG-SNS1Y	N1-NetEngine 8000 F8 BNG Function License SnS	
N1-N8KF8MED-SNS1Y	N1-NetEngine 8000 F8 Media Fabric Function License SnS	
N1-N8KF8SRV6D1-SNS1Y	N1-NetEngine 8000 F8 G-SRv6 32bits SRH Compression Function License SnS	
N1-N8KF8SRV6D2-SNS1Y	N1-NetEngine 8000 F8 G-SRv6 16bits SRH Compression Function License SnS	
N1-N8KF8ESF1D1-SNS1Y	N1-NetEngine 8000 F8 Energy-Saving Enhanced Function Package 1.0 SnS	
N1-N8KF8ZTA-SNS1Y	N1-NetEngine 8000 F8 Zero Trust Access Function License SNS	
N1-N8KF8HIPS-SNS1Y	N1-NetEngine 8000 F8 Host-based Intrusion Prevention System Function License SNS	
N1-N8KF8CC-SNS1Y	N1-NetEngine 8000 F8 Configuration Check Function License SNS	
N1-N8KF8TEE-SNS1Y	N1-NetEngine 8000 F8 Security Enhanced Trusted Execution Environment Function License SNS	

Optical/Electrical Transceivers

100Gbps Optical Transceiver

Order Name	Description	Remarks
OMND10N13	High Speed Transceiver,QSFP28,850nm,100G, -8.4dBm,2.4dBm,-10.3dBm,MPO,MM,0.1km,OM4	
OSN020N15	High Speed Transceiver,QSFP28,1310nm,4*25GBase, -6.5dBm,2.5dBm,-9.8dBm,LC/PC,2km (100GBASE- CWDM4)	
OSN010N24	High Speed Transceiver,QSFP28,1310nm,100G, -4.3dBm,4.5dBm,-8.6dBm,LC,10km	
QSFP28-100G- 1310-40km-SM	High Speed Transceiver,QSFP28,1310nm,100G, -2.9dBm,2.9dBm,-20.9dBm,LC,40km	
QSFP28-100G- 1310-80km-SM	High Speed Transceiver,QSFP28,1310nm,100G,3dBm, 6.5dBm,-30dBm,LC,SMF,80km	

50Gbps Optical Transceiver

Order Name	Description	Remarks
OSL010N01	High Speed Transceiver,QSFP28,1311nm,50Gb/s,-4dBm, 4.2dBm,-8.8dBm,LC,SMF,10km	
QSFP28-50G- 1311nm-40km-SM	High Speed Transceiver,QSFP28,1310nm,53.125G,1.5dBm, 8dBm,-15dBm,Stright LC,SMF,40km	
QSFP28-50G- 1310nm-80km-SM	High Speed Transceiver,QSFP28,1311nm,50G,two-fiber bidirectional,80km	

40Gbps Optical Transceiver

Order Name	Description	Remarks
OMXD30009	Optical Transceiver,QSFP+,1310nm,41.25Gbps,-7dBm, 2.3dBm,-11.5dBm,LC,SMF,10km	
OMXD30010	High Speed Transceiver,QSFP+,850nm,41.25Gbps, -7.6dBm,0.5dBm,-9.5dBm,MPO,MMF,0.15km	

10Gbps Optical Transceiver

Order Name	Description	Remarks
OMXD30002	Optical Transceiver(SFP+,1550nm,9.95~11.1Gb/s,-4.7~4dBm, -14.1dBm,LC,SM,40km)	
OSG080N01	Optical Transceiver,eSFP,1550nm,1.25Gb/s,-2~5dBm, -23dBm,LC,SM,80km	
OSX040N03	Optical Transceiver(SFP+,850nm,10Gb/s,-7.3~-1dBm, -11.1dBm,LC, MM,0.3km)	
OSX001002	Optical Transceiver(SFP+,1310nm,10Gb/s,-8.2~0.5dBm, -12.6dBm,LC,SM,10km)	
OSX080N04	Optical Transceiver(SFP+,1550nm,9.953~10.3125Gb/s, 0~4dBm,-24dBm,LC,SM,80km)	
OSX010N13	XFP SFP+ Transceiver-SFP+-1310nm-10G/1G--8.2dbm-0.5dbm-Single-mode-14.4dbm-LC-10km	

1Gbps Optical Transceiver

Order Name	Description	Remarks
OMGD50N02	Optical Transceiver(eSFP,850nm,2.125Gb/s(Multi rate),-9.5~-2.5dBm,-17dBm,LC,MM,0.5km)	
OSG010N05	Optical transceiver,eSFP,1310nm,1.25Gb/s,-9~-3dBm, -20dBm,LC,Single Mode,10km	
OSG040N02	Optical Transceiver(eSFP,1310nm,1.25Gb/s,-5~0dBm,-23dBm,LC,SM,40km)	
OSG080N01	Optical Transceiver,eSFP,1550nm,1.25Gb/s,-2~5dBm,-23dBm,LC,SM,80km	

1 Gbps Electrical Transceiver

Order Name	Description	Remarks
OEGD01N02	Optical functional module,1000BASE-T -SFP RJ45Electrical Module,Auto Negotiate,100m ,-40~85degC	

622M SFP/eSFP Optical Transceiver

Order Name	Description	Remarks
eSFP-1310nm-L-4.1	Optical Transceiver(eSFP,1310nm,STM4,-3dBm~2dBm,-30dBm,Singlemode,LC,40km)	
OSH080N01	Optical Transceiver,NetEngine40E,CR5M1SF21,Optical Transceiver,eSFP,1550nm,STM4,-3~2dBm,-30dBm,LC,SM,80km	

155M SFP/eSFP Optical Transceiver

Order Name	Description	Remarks
eSFP-FE-LX-SM1310	Optical Transceiver(eSFP,1310nm,STM1,-15dBm~-8dBm,-31dBm,Singlemode,LC,15Km)	
eSFP-FE-LH40-SM1310	Optical Transceiver(eSFP,1310nm,STM1,-5dBm~0dBm,-37dBm,Singlemode,LC,40Km)	

BIDI Optical Transceiver

Order Name	Description	Remarks
SFP-GE-LX-SM1310-BIDI	Optical Transceiver,ESFP,Tx1310/Rx1490,1.25Gb/s,-3dBm,-9dBm,-19.5dBm,LC,Single Mode,10km	
SFP-GE-LX-SM1490-BIDI	Optical Transceiver,ESFP,Tx1490/Rx1310,1.25Gb/s,-3dBm,-9dBm,-19.5dBm,LC,Single Mode,10km	
OGEBIDI41	Optical Transceiver,eSFP,1310nm(Tx)/1490nm(Rx),1.25Gb/s,-2dBm,3dBm,-23dBm,LC,SM,40km	
OGEBIDI40	Optical Transceiver,eSFP,1490nm(Tx)/1310nm(Rx),1.25Gb/s,-2dBm,3dBm,-23dBm,LC,SM,40km	
OSX010B10	BiDi Transceiver,SFP+,1270nm(Tx)/1330nm(Rx),2.5~11.3Gb/s,-8.2dBm,0.5dBm,-14.4dBm,LC,SM,10km	
OSX010B11	BiDi Transceiver,SFP+,1330nm(Tx)/1270nm(Rx),2.5~11.3Gb/s,-8.2dBm,0.5dBm,-14.4dBm,LC,SM,10km	
QSFP28-50G-BIDI-10km-SM-1	BiDi Transceiver,QSFP28,1331nm(TX)/1271nm(RX),50G,-4.5dBm,4.2dBm,-8.9dBm,LC,10km	

Order Name	Description	Remarks
QSFP28-50G-BIDI-10km-SM-2	BiDi Transceiver, QSFP28, 1271nm(TX)/1331nm(RX), 50G, -4.5dBm, 4.2dBm, -8.9dBm, LC, 10km	
QSFP28-50G-BIDI-40km-SM-1	BiDi Transceiver, QSFP28, 1309nm(TX)/1295nm(RX), 50G, 1.5dBm, 8dBm, -13.5dBm, LC, SMF, 40km	
QSFP28-50G-BIDI-40km-SM-2	BiDi Transceiver, QSFP28, 1295nm(TX)/1309nm(RX), 50G, 1.5dBm, 8dBm, -13.5dBm, LC, SMF, 40km	

For More Information

For more information about the NetEngine 8000 F8, visit <https://e.huawei.com> or contact us in the following ways:

- Global service hotline: <https://e.huawei.com/en/service-hotline>
- Logging into the Huawei Enterprise technical support web: <https://support.huawei.com/enterprise/>
- Sending an email to the customer service mailbox: support_e@huawei.com

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